

An Experimental Study of Neural Clause Selection*

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- important heuristic point in saturation-based theorem proving
- concerned with: Which clause to activate next?
- favorite target for ML: prefer clauses that appeared in past successful proofs

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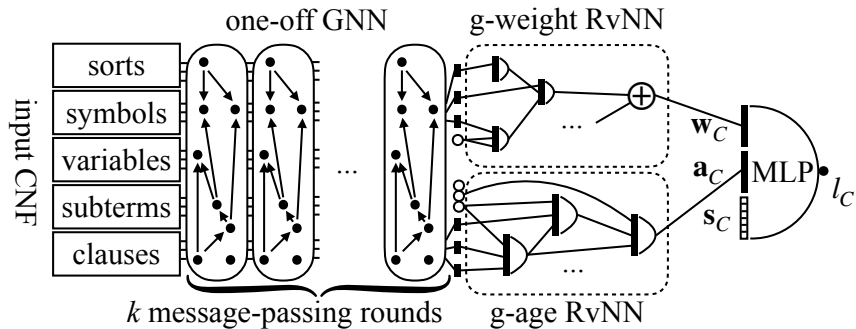
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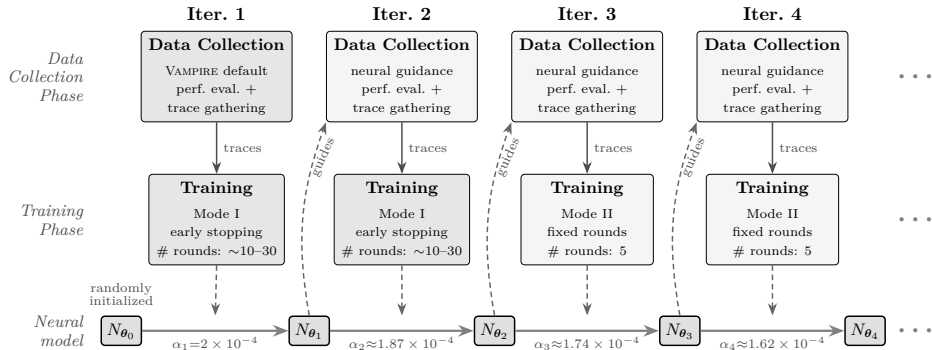
Trace: Every Moment Counts!

$$T = (P_T, \mathcal{C}, \mathcal{C}^+, \{\mathcal{P}_i\}_{i \in I_T})$$

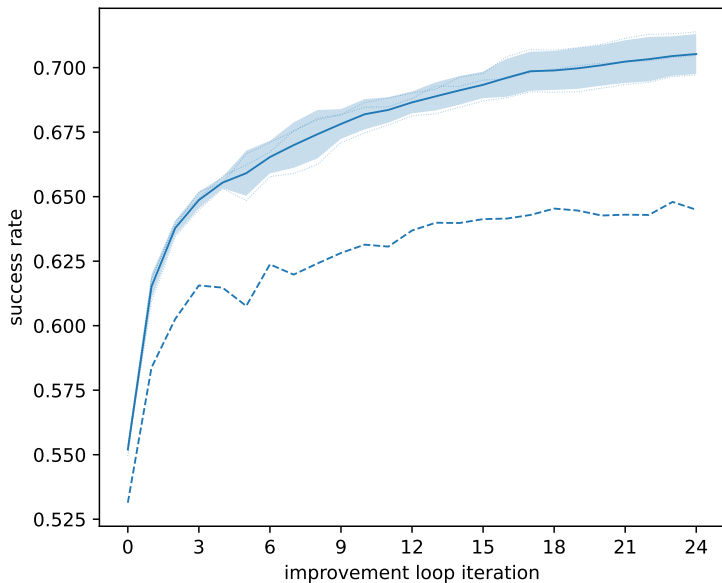
Neural Architecture Diagram



Iterative Improvement Loop



Typical Performance Evolution (train: +27.76%, test: +21.93%)

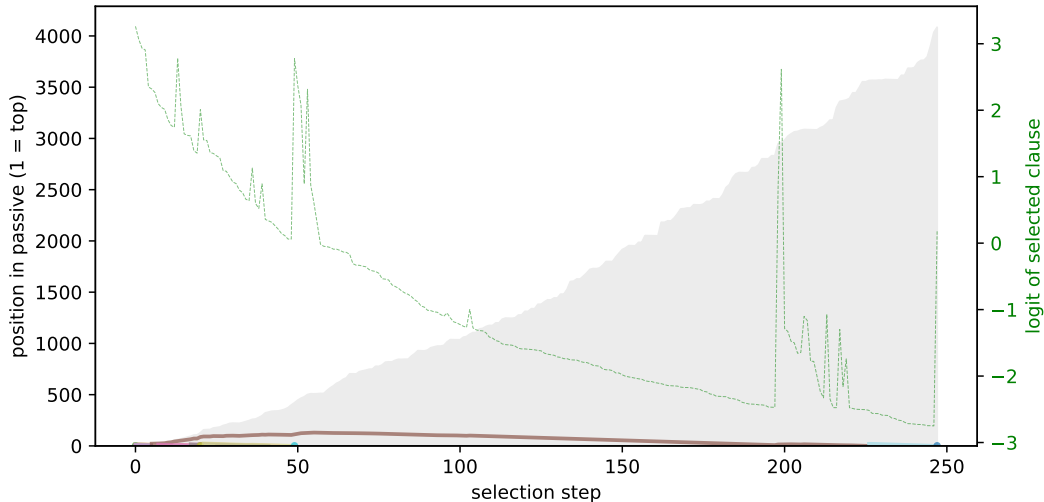


Solving REL024-2 (first solved in iteration 2)

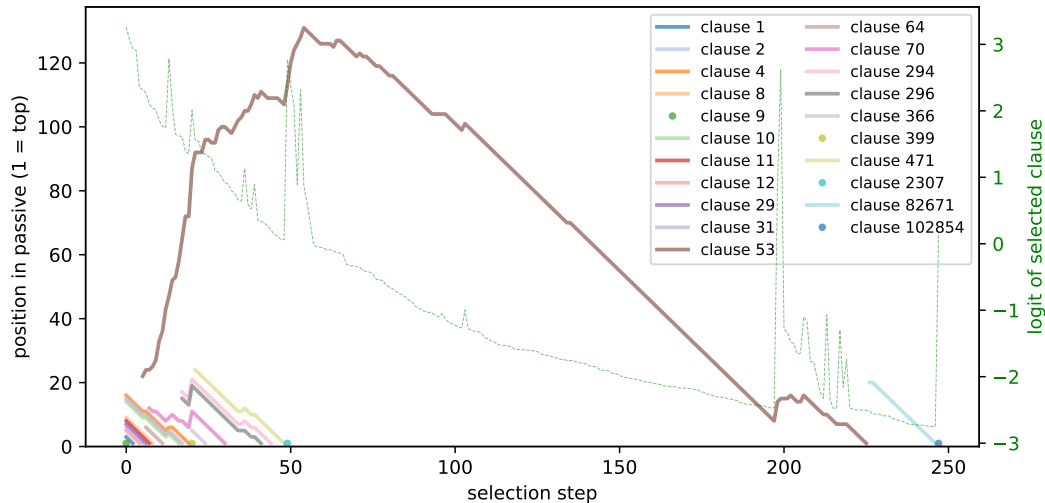
A simple consequence of isotonicity (Relation-Algebraic Semantics); UEQ; rating 0.26

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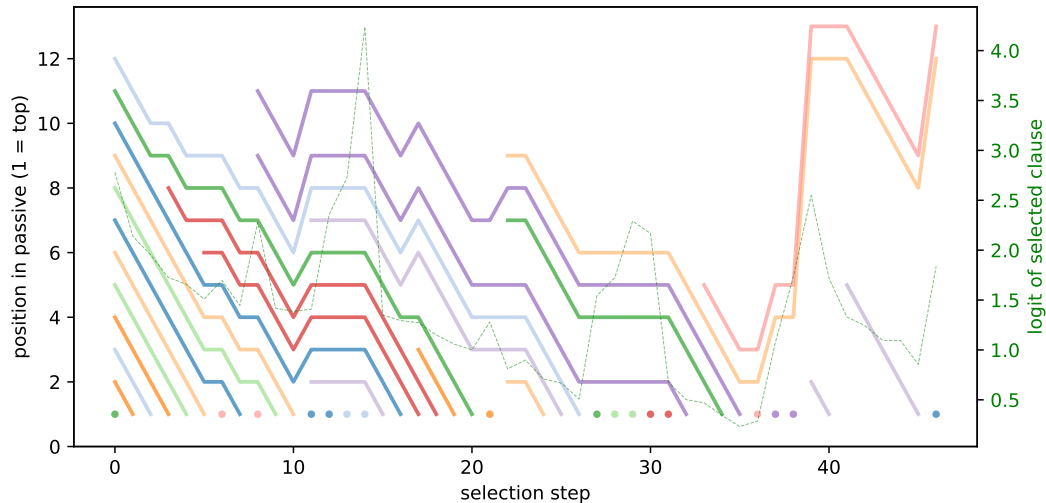
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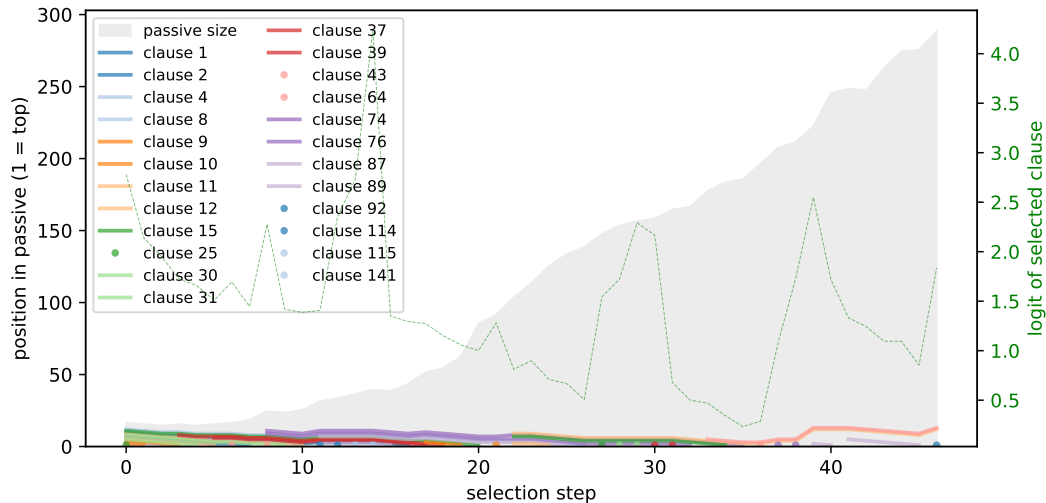
Solving REL024-2 (zooming in on the interesting part)



Solving REL024-2 (after 30 iterations of training)



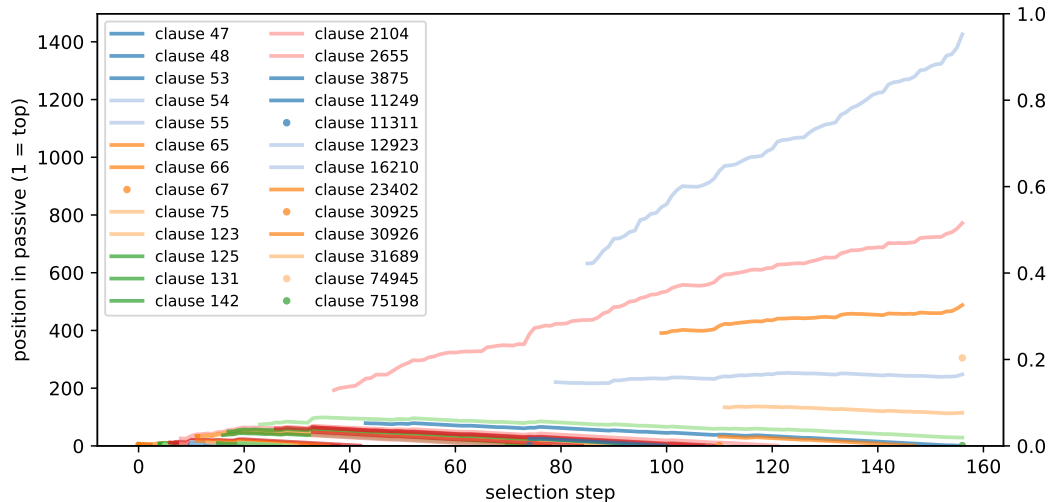
Solving REL024-2 (after 30 iterations; zooming out again)



$$(X \cap Y) \cup (Y \cap Z) \cup (Z \cap X) = (X \cup Y) \cap (Y \cup Z) \cap (Z \cup X) ; \text{FOF; rating 0.39}$$

Solving SET601+3

$(X \cap Y) \cup (Y \cap Z) \cup (Z \cap X) = (X \cup Y) \cap (Y \cup Z) \cap (Z \cup X)$; FOF; rating 0.39



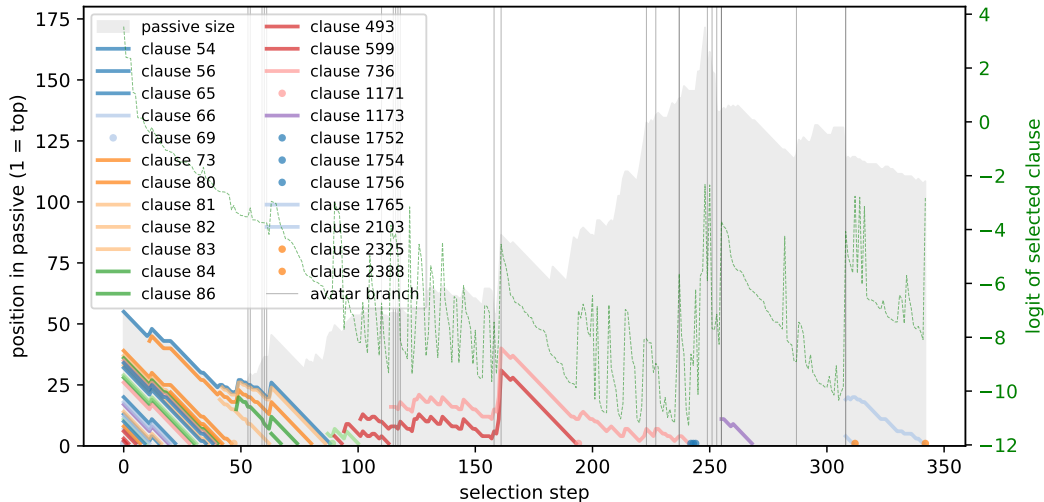
Solving SET601+3 (the end of the proof)

```
.
.
.
73013. union(intersection(X2,X1),intersection(union(intersection(X0,intersection(X2,X1)),X1),X3)) =
        intersection(union(X0,X1),union(intersection(X2,X1),intersection(X1,X3))) [superposition 30926,3875]
73719. intersection(union(X2,X1),union(X3,intersection(X0,X1))) =
        union(intersection(X1,X0),intersection(union(intersection(X2,X3),X1),X3)) [forward demodulation 72949,31689]
73733. union(intersection(X2,X1),intersection(union(intersection(X0,intersection(X2,X1)),X1),X3)) =
        intersection(X1,intersection(union(X0,X1),union(intersection(X2,X1),X3))) [forward demodulation 73013,2655]
74249. intersection(union(X2,X1),union(X3,intersection(X0,X1))) =
        union(intersection(X1,X0),intersection(X3,union(intersection(X2,X3),X1))) [forward demodulation 73719,318]
74262. union(intersection(X2,X1),intersection(union(intersection(X0,intersection(X2,X1)),X1),X3)) =
        intersection(union(X0,X1),intersection(union(X3,X2),X1)) [forward demodulation 73733,23402]
74645. intersection(union(X2,X1),union(X3,intersection(X0,X1))) =
        union(intersection(X1,X0),union(intersection(X2,X3),intersection(X3,X1))) [forward demodulation 74249,288]
74656. union(intersection(X2,X1),intersection(union(intersection(X0,intersection(X2,X1)),X1),X3)) =
        intersection(X1,intersection(union(X0,X1),union(X3,X2))) [forward demodulation 74262,131]
74945. intersection(union(X2,X1),union(X3,intersection(X0,X1))) =
        union(intersection(X2,X3),intersection(X1,union(X3,X0))) [forward demodulation 74645,16210]
74956. union(intersection(X2,X1),intersection(union(intersection(X0,intersection(X2,X1)),X1),X3)) =
        intersection(X1,union(X3,X2)) [forward demodulation 74656,1308]
75198. intersection(X1,union(X3,X2)) =
        union(intersection(X2,X1),intersection(X3,X1)) [forward demodulation 74956,12923]
75885. union(intersection(sk4,sk2),intersection(sk3,union(sk2,sk4))) !=
        intersection(union(sk4,sk3),union(sk2,intersection(sk4,sk3))) [superposition 2104,75198]
76058. $false [forward subsumption resolution 75885,74945]
```

Semantic Web (Cliques); FOF; rating 0.30

Solving SWB013+2

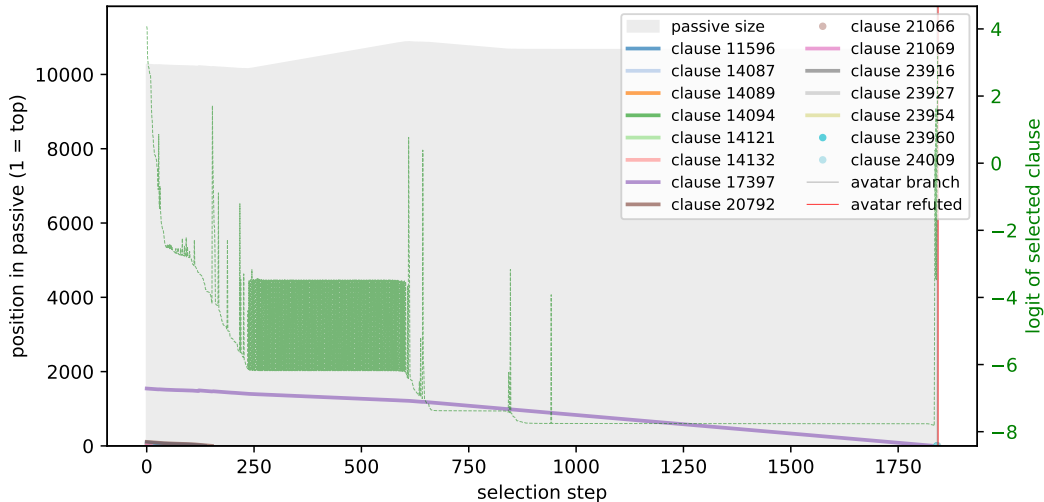
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Commonsense Reasoning (Where can you find the Statue of Liberty?); FOF; rating 0.35

Solving CSR113+4

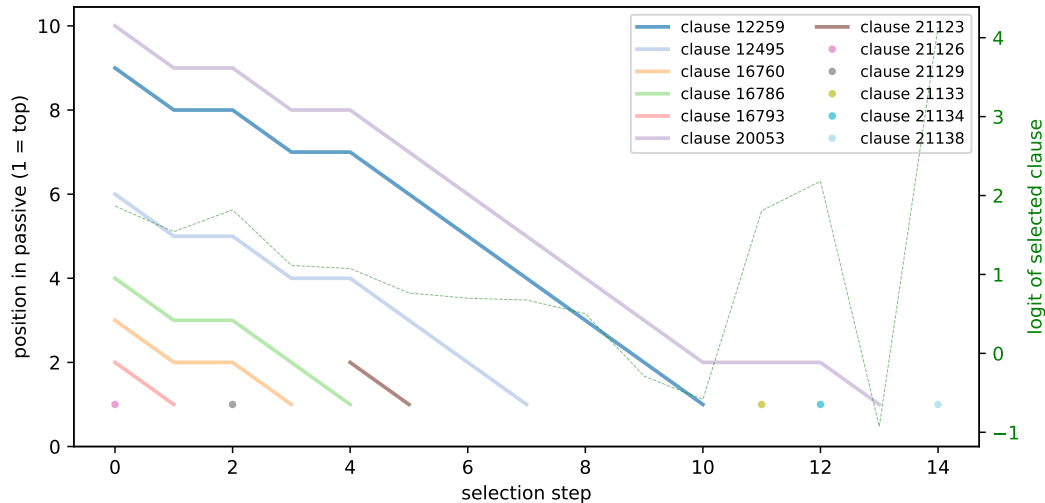
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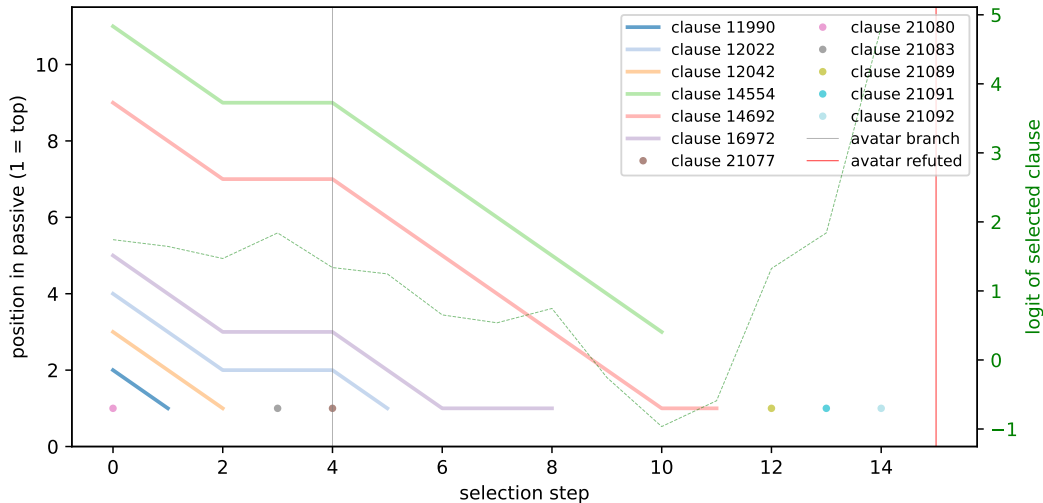
Solving CSR113+4 (What the hack is happening in the 'green box'?)

```
...
379 [SA] active: 16103. state_adjective_state_binding(guinea_bissauisch_1_1,guinea_bissau_0) [cnf transformation 9054]
380 [SA] forward reduce: 21370. ~prop(X0,guinea_bissauisch_1_1) | loc(X0,sk27(X0,guinea_bissau_0)) [resolution 16103,14618]
381     replaced by 21371. ~prop(X0,guinea_bissauisch_1_1) <- (2) [forward subsumption resolution 21370,21128]
382     using 21128. ~loc(X0,X1) <- (2) [resolution 18250,21070]
383 [SA] active: 21371. ~prop(X0,guinea_bissauisch_1_1) <- (2) [forward subsumption resolution 21370,21128]
384 [SA] active: 16204. state_adjective_state_binding(antiguanisch_1_1,antigua_und_barbuda_0) [cnf transformation 9009]
385 [SA] forward reduce: 21377. loc(X0,sk27(X0,antigua_und_barbuda_0)) | ~prop(X0,antiguanisch_1_1) [resolution 16204,14618]
386     replaced by 21378. ~prop(X0,antiguanisch_1_1) <- (2) [forward subsumption resolution 21377,21128]
387     using 21128. ~loc(X0,X1) <- (2) [resolution 18250,21070]
388 [SA] active: 21378. ~prop(X0,antiguanisch_1_1) <- (2) [forward subsumption resolution 21377,21128]
389 [SA] active: 11827. state_adjective_state_binding(indonesisch_1_1,indonesien_0) [cnf transformation 9061]
390 [SA] forward reduce: 21384. loc(X0,sk27(X0,indonesien_0)) | ~prop(X0,indonesisch_1_1) [resolution 11827,14618]
391     replaced by 21385. ~prop(X0,indonesisch_1_1) <- (2) [forward subsumption resolution 21384,21128]
392     using 21128. ~loc(X0,X1) <- (2) [resolution 18250,21070]
393 [SA] active: 21385. ~prop(X0,indonesisch_1_1) <- (2) [forward subsumption resolution 21384,21128]
394 [SA] active: 19018. state_adjective_state_binding(belgisch_1_1,belgien_0) [cnf transformation 9018]
395 [SA] forward reduce: 21391. ~prop(X0,belgisch_1_1) | loc(X0,sk27(X0,belgien_0)) [resolution 19018,14618]
396     replaced by 21392. ~prop(X0,belgisch_1_1) <- (2) [forward subsumption resolution 21391,21128]
397     using 21128. ~loc(X0,X1) <- (2) [resolution 18250,21070]
398 [SA] active: 21392. ~prop(X0,belgisch_1_1) <- (2) [forward subsumption resolution 21391,21128]
399 [SA] active: 20497. state_adjective_state_binding(swasil__344ndisch_1_1,swasiland_0) [cnf transformation 9159]
400 [SA] forward reduce: 21398. ~prop(X0,swasil__344ndisch_1_1) | loc(X0,sk27(X0,swasiland_0)) [resolution 20497,14618]
401     replaced by 21399. ~prop(X0,swasil__344ndisch_1_1) <- (2) [forward subsumption resolution 21398,21128]
402     using 21128. ~loc(X0,X1) <- (2) [resolution 18250,21070]
403 [SA] active: 21399. ~prop(X0,swasil__344ndisch_1_1) <- (2) [forward subsumption resolution 21398,21128]
404 [SA] active: 14437. state_adjective_state_binding(liechtensteinisch_1_1,liechtenstein_0) [cnf transformation 9094]
405 [SA] forward reduce: 21405. loc(X0,sk27(X0,liechtenstein_0)) | ~prop(X0,liechtensteinisch_1_1) [resolution 14437,14618]
406     replaced by 21406. ~prop(X0,liechtensteinisch_1_1) <- (2) [forward subsumption resolution 21405,21128]
407     using 21128. ~loc(X0,X1) <- (2) [resolution 18250,21070]
408 [SA] active: 21406. ~prop(X0,liechtensteinisch_1_1) <- (2) [forward subsumption resolution 21405,21128]
...
```

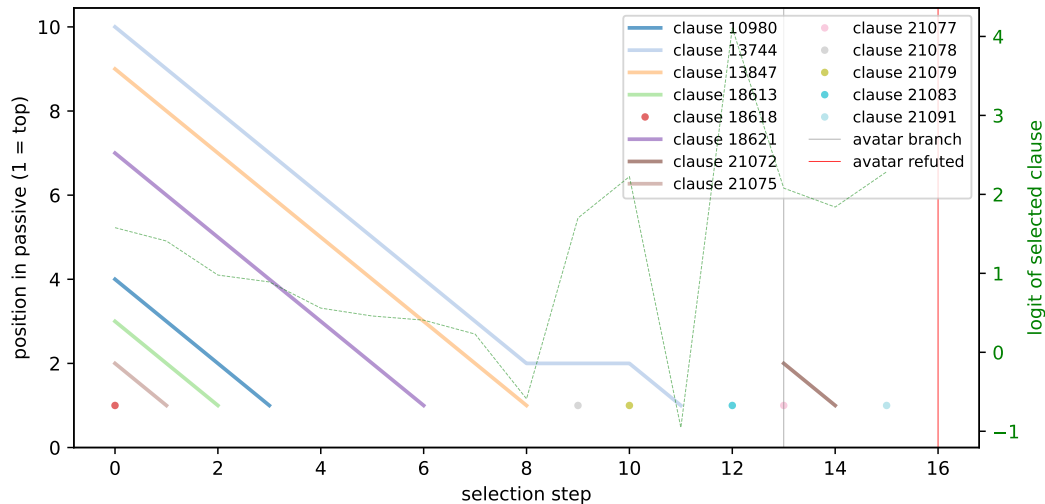
Solving CSR113+4 (thirty iterations later)



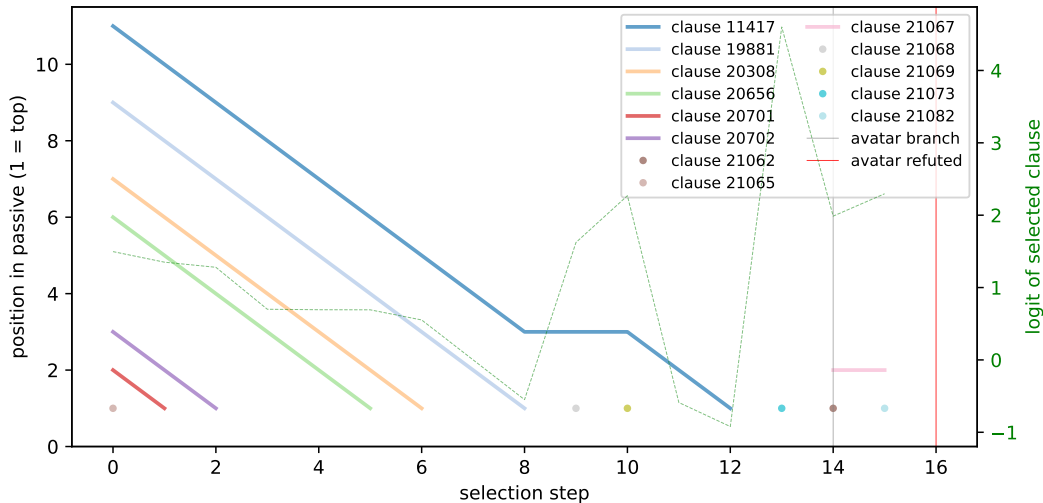
Solving CSR113+4 (thirty iterations later; under different random seeds)



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Thank you!

Solving SYO687-1

Syntactic (GradedStrictMonotoneSequenceSchema from GAPT); CNF; rating 0.31

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